SOURABH PORWAL

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Functional safety engineer with seven years in automotive and autonomous driving, currently owning functional safety for Vay's remote-driving system, a fleet of around 200 cars running on public roads in Las Vegas with no one in the vehicle. Day to day that means owning the safety requirements and the safety architecture, writing system, subsystem, and component requirements for individual features, and working with the test teams to verify them. Hands-on across the full ISO 26262 lifecycle, work closely with the OEM and partner teams, and now lead the safety work as Vay brings AI into the driving task itself.

SKILLS & TOOLS

Functional Safety: ISO 26262, ISO 21448 (SOTIF); HARA, FMEA, FTA, FMEDA; FSC / TSC, safety case

Systems & Requirements Engineering: Requirements engineering & traceability (JAMA, DOORS); V&V at HiL and vehicle level; V-model development

AD Standards & Regulation: ISO 26262, UNECE R 157, ISO 4804, FMVSS, UL 4600, ISO/IEC TS 5083

AI & Autonomy: Safety of AI/ML models in safety-critical driving functions; remote-driving operations on public roads

Programming: C / C++, Python (general working knowledge)

EXPERIENCE

Senior Functional Safety Engineer
09/2023 – Present

Vay Technologies GmbH · Berlin, Germany

- Own functional safety for Vay’s remote-driving system, driving ISO 26262 compliance across the systems, hardware, and software teams from concept through real-world operation on public roads.
- Own and author the safety case (UL 4600), and define the functional and technical safety architecture, including system-level redundancies.
- Define the interface between the safety controller and the vehicle over CAN and LIN, analyse failure modes across the system, and specify the corresponding system reactions.
- Derive tolerable failure-rate thresholds for each fallback-manoeuvre type (MRM) from probabilistic risk analysis, with an operational stop-fleet trigger if the thresholds are exceeded.
- Lead the safety work for bringing AI into safety-critical driving functions.
- Act as safety lead for integrating the remote-driving stack into a B2B customer’s autonomous-driving product as a fallback layer, working directly with the partner’s safety and engineering teams on architecture, integration, and safety deliveries.

Functional Safety Engineer II
06/2022 – 08/2023

Vay Technologies GmbH · Berlin, Germany

- Designed the system’s fail-safe fallback behaviour (Minimum Risk Manoeuvre) end to end, from concept and systems through functional safety, applying ISO 26262, UNECE R 157, and ISO 4804.
- Led the functional-safety side of the independent TÜV SÜD assessment of the safety concept and risk analyses, contributing to Vay’s first-in-Europe approval to operate on public roads without a safety driver.
- Performed safety analyses (FMEA, FTA) and supported verification and validation at system and vehicle level.

Functional Safety Engineer I
07/2021 – 06/2022

Vay Technologies GmbH · Berlin, Germany

- Helped establish the in-house functional safety function and processes from scratch.
- Implemented safety requirements management and traceability in JAMA.

- Contributed to the company's first successful external safety assessment.

Functional Safety Engineer / Manager 01/2020 – 06/2021
BERNS Engineers GmbH · Gilching, Germany

- Prepared safety concepts and analyses (FSC, TSC, HARA, FTA) for OEM customers and worked directly with their architects on ASIL D ADAS features.
- As functional safety manager on certain projects, owned the safety plan, oversaw supplier safety activities, and provided functional-safety releases for the OEM's internal development milestones.

Functional Safety Engineer 06/2019 – 12/2019
AKKA Technologies · Ingolstadt, Germany

- Supported ISO 26262 development for an ASIL B system and managed vehicle configuration data.

Junior Executive 08/2013 – 09/2015
Fleetguard Filters Pvt. Ltd. · Pune, India

- Managed production part approval process (PPAP) activities across manufacturing suppliers, including process FMEA and process flow documentation; reduced component cost by ~25% through supplier negotiations.

EDUCATION

M.Sc. Mechanical Engineering 10/2015 – 04/2019
Rhine-Waal University of Applied Sciences · Kleve, Germany

B.E. Mechanical Engineering 08/2009 – 04/2013
Visvesvaraya Technological University · Belgaum, India

CERTIFICATION

CertX – ISO 26262:2018 Functional Safety Red Belt (03/2021)

PERSONAL DETAILS

Nationality: German
Languages: English (fluent), German (B1), Hindi (native)